

Laboratory full-scale test on the bond property of GFRP anchor to concrete

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ABSTRACT

Glass fiber reinforced polymer (GFRP) bolt, which has excellent mechanical properties and corrosion resistance, is gradually replacing metal bolt in rock and soil anchoring, compared with traditional reinforced bolts and other composite materials with high cost. In the current study, the influence of anchorage length and anchorage mode on the bonding property of bolt and concrete was investigated, and the stress mechanism of glass fiber bolt was analyzed based on the tensile test of a full-size laboratory sample. The results show that the failure modes of GFRP anti-floating anchor were pull-out failure and fracture failure, the ultimate bearing capacity of mechanical anchorage was noted as 1.08–1.24 times that of linear anchorage. Under the same anchorage method, with the increase of anchorage length, the relative displacement and the proportion of load bearing in the easing section of the bolt load–displacement curve decreased; thus, the load–displacement curve of the anchor in the form of mechanical anchorage was smoother. With the increase of load, the main load bearing mode of the bolt-concrete interface changed from the interface friction force and chemical adhesive force to the mechanical bite force, and then the interface concrete was broken, and the concrete no-acting zone was transformed into the acting zone until the bolt was damaged. Anchor had a maximum anchorage length beyond which the bearing capacity of the anchors increased minimally. Comparing the calculation formulas of anchor length under different national standards, it was suggested to use Chinese or European standards to calculate anchor length.

1. Introduction

Anti-float engineering, an important rock, and soil anchorage branch is more commonly used in coastal city construction, particularly underground structures. The anti-floating treatment of structures has become essential in recent years with the growth of urban underground buildings. Due to its easy-to-use design and affordable cost, the anti-floating bolt is widely applied. The bond between concrete and reinforcement is one of the key points in designing an anti-floating anchor; as a common anti-floating anchor reinforcement, steel bars are increasingly replaced by glass fiber reinforced polymer (GFRP) with high tensile strength and strong corrosion resistance [1,2]. The bonding mechanism of GFRP bars differs from traditional steel bars due to the varied materials and mechanical properties [3].

As a long-term support, the material and mechanical properties of the GFRP anti-floating anchor structure have been studied comprehensively. Essentially, the chemical bond, along with bearing resistance and friction resistance, contributes particularly to the bond strength between the reinforcement body and the concrete [4]. With the increase in

temperature, the bonding strength between concrete and reinforcement decreases [5,6,7], and GFRP reinforcement is also affected by temperature to produce a large deformation [7]. Gooranorimi et al. [8] proposed concrete splitting and pull-out failure modes through the bond slip model, which simulated the failure forms of anchor solids, found that the thickness of concrete would affect the bonding strength of bolt. Kou et al. [9] proved that the GFRP anchorage system had a critical depth through embedded FBG sensing technology, and beyond this critical depth, the bearing capacity of the structure would not be affected by the depth. In another study performed by Li et al. [10], similar conclusions were obtained through experiments. The bolt sample with low concrete strength or short bond length has a large slip [11,12,13], and under continuous load, the bond stress of the sample with a long bond length varies considerably [11]. The GFRP bars with larger diameter or shorter anchorage length have higher bond strength, as demonstrated by the experiments of Yoo et al. [14] and Tastani et al. [15]. According to Fava et al. [16]'s finite element simulation, steel bars have a better bonding effect than GFRP bars. Under cyclic load, the bond strength of GFRP bars deteriorates more seriously than that of steel bars

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[17]. Even if the GFRP bolt is subjected to approximately 80% of the ultimate tensile strength and micro-cracks occur in the rod body, its durability will not be affected [18]. Furthermore, even if it is submerged in a salt solution for a long duration, its durability won't be significantly affected [19,20]. Under long-term load, GFRP bars with low fiber reinforcement content and large diameter have higher creep level, and the longitudinal tensile properties of the bars are unchanged [21].

Anchorage length is one of the essential parameters for the anti-floating design of an anchor, and its value directly affects the bonding performance between the anchor and anchor solid. The strength of the bolt can be completely exploited, and the project's quality is ensured by using reasonable anchoring length. However, research on the anchorage length of the anti-floating bolt is limited in number. The relevant regulations on the anchorage length of the metal bolt are not precise and are generally conservative, which is easy to cause material waste and does not apply to GFRP bolt with high tensile strength. In view of this, through the laboratory test of pulling destructiveness of GFRP anchor, exploring the influence of anchorage length and anchorage mode on the bonding property between GFRP anchor and concrete in the current study. The critical value of anchorage length was analyzed. A comparison of the anti-floating bolt anchorage length calculation methods used in various national codes and a discussion of their applicability to GFRP anchorage length calculations were conducted.

2. Mechanical properties of GFRP anchor

Composite material is a new material that combines two or more different materials to obtain an overall performance higher than that of a single component [22]. Fiber-reinforced polymer (FRP) is a composite material composed of fiber (load bearing) and matrix (bonding material, distributed load, protective fiber). Aramid, carbon, and glass fibers are commonly used in civil engineering applications. Among these, aramid fiber exhibits good fatigue resistance, carbon fiber shows good fatigue resistance and high longitudinal tensile strength, and glass fiber has relatively wide-ranging mechanical and physical properties. Table 1 explicitly indicates the mechanical properties of these fibers [23]. The matrix serves as a binder to keep the fibers in place and provides the composite with structural stability by enabling shear transfer. Compared with traditional rebar, the advantages of composite materials are lightweight, high tensile strength, and good durability [24].

The high fatigue resistance, toughness, high-temperature wear, and oxidation resistance of GFRP bolts, along with their capacity to be manufactured with tailored strength properties, make them an excellent option for anti-floating engineering applications [25]. Moreover, it stands out among many FRP bolts due to its low cost. It is made of glass fiber as reinforcement, with good thermal stability and chemical resistance, and undergoes low creep and stress relaxation thermosetting resin as the matrix (including epoxy resin, polyester, vinyl ester, etc.) through the extrusion process. The glass fiber is impregnated with resins and additives using an extruder in a dip tank, and the wet fiber is then cured in a mold. After forming, the surface is generally machined into a threaded shape.

The density of the GFRP anchor rod is only 1/4–1/5 of that of a steel bar; however, its tensile strength is greater than that of the steel bar [26]; hence, due to the high tensile strength, good acid and alkali

corrosion resistance, strong electromagnetic interference resistance, high-cost performance, fast construction speed, and other advantages, the underlying issue of metal bolts' poor corrosion resistance can be resolved by GFRP bolts, which can also improve the structure's durability and strengthen the structure's safety. On the other hand, the disadvantage is that the compressive and shear strength is low and cannot be welded or bent by construction technology. Depending on the material properties and forming process of the GFRP bolt, it is not easy to decompose in the soil layer in the process of use, thus preventing soil contamination. Regarding the related feature, it can be stated as a novel type of environment-friendly rock and soil anchorage material.

3. Laboratory model test of GFRP anti-floating anchor

3.1. Test purpose

At present, most scholars have studied the bond property of GFRP bolt and cement mortar, yet tests on the bond property of GFRP anti-float bolt and concrete are limited in number. As an important index of anchorage performance of anti-floating anchors, the value of anchorage length greatly influences the safety of anti-floating engineering. In this paper, through the laboratory pulling model test of the linear anchor bolt and bolt with mechanical anchorage, the influence of anchoring length and mode on the GFRP anti-floating anchor's anchoring performance was studied.

3.2. Test design

The single root group and horizontal pull test method were adopted to reflect the actual working conditions and ensure the scientific conclusion. Specific methods were as follows: A anchor was connected in series with two anchor solids. The entire body formed by the bolt bonded with the soil through cement slurry or cement mortar was referred anchor solid, and the length of the bolt in the anchor solid was called anchor length. Concrete was used to simulate the anchor solids and act as the foundation plate. Two sets of perforating hydraulic jacks were arranged horizontally on both sides of the bolt to prevent the large local force from affecting the test results. During installation, it was ensured that the jack and GFRP anchor shaft were in the same horizontal plane, and the axial tension of the GFRP anchor was ensured. A displacement dial indicator was installed at the contact surface of the GFRP bolt and anchor solid to measure the slip of the GFRP bolt directly, the dial indicator was 50 mm from the bolt. Four displacement dial gauges were symmetrically installed on both sides of the tension specimen at the same elevation as the GFRP bolt. These gauges were utilized to monitor the oil pressure value of the two jacks to ensure the axial tension of the GFRP bolt. The problem that the GFRP anchor has high requirements for the jig due to its weak shear performance is eliminated since there is no need to use an anchor head jig in comparison to the traditional anchor-pulling device. Horizontal loads were applied to anchor solids by two sets of center hole jacks. By applying a load during the test process, two sets of parallel tests could be completed, and two sets of test data could be obtained.

Two kinds of anchor solids ($800 \times 800 \times 500$ mm and $800 \times 800 \times 900$ mm) were used to test GFRP anchors with different anchorage lengths. The test device is illustrated in Fig. 1 (As an example, the anchor solid size of $800 \times 800 \times 500$ mm is taken).

3.3. Test materials

3.3.1. GFRP anti-floating anchor

The GFRP bolt used in the test had a sticky sand surface, in which epoxy resin accounted for approximately 25%, and glass fiber accounted for about 75%. According to the relevant parameters provided by the manufacturer, the density of the bolt was 2.1 g/cm^3 , the average ultimate tensile strength of the bolt was 432 kN, the ultimate tensile

Table 1
Mechanical parameters of metal and fiber bolt.

	Nominal yield stress MPa	Tensile strength MPa	Elastic modulus $\times$ 10^3 MPa	Yield strain (%)	Ultimate strain (%)
Steel	276–517	482–689	200	1.4–2.5	6–12
GFRP	NA	482–1585	35–51	NA	1.2–3.1
CFRP	NA	600–3688	103–579	NA	0.5–1.9
AFRP	NA	1724–2537	41–125	NA	1.9–4.4

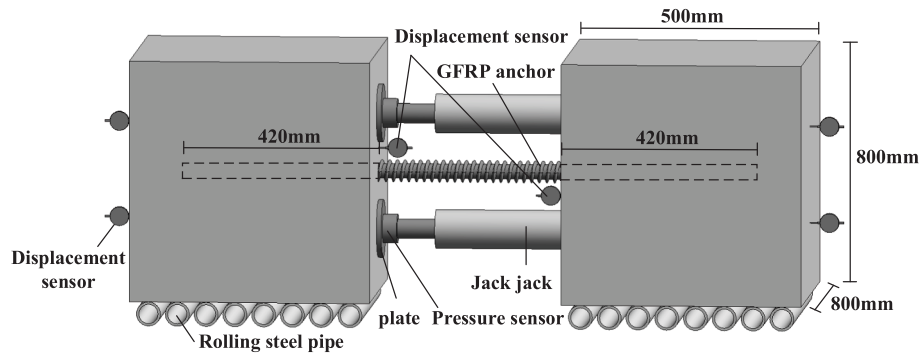


Fig. 1. Schematic diagram of the test device.

strength was 702 MPa, the ultimate shear strength was 150 MPa, and the elastic modulus was 51 GPa. The bolt diameter was 28 mm, the bolt length was 420 mm (15D) and 840 mm (30D), respectively, and D was the bolt diameter. The authors developed and customized mechanical anchors, as demonstrated in Fig. 2. The mechanical anchor was made of the same GFRP material, composed of an anchor disc and a conical nut. The diameter of the anchor disc in the mechanical anchor was 170 mm, the thickest part of the axis was 10 mm, the inner diameter of the cone nut was 28 mm, and the axial length was 60 mm. The ultimate bearing capacity of the mechanical anchor was determined as 120 kN. Test anchor parameters are presented in Table 2.

3.3.2. Anchor solid

The anchor solid in the test was plain concrete with a strength grade of C25. The sectional dimension was 800 × 800 mm, and the thicknesses were 500 mm and 900 mm. Anchor solids are made by making box formwork and then pouring concrete. Seamless steel tubes were positioned at the bottom of the container to reduce ground friction during the test loading process, which could alter the test results due to the uneven force of the anchor solids. The test bolt hole, set aside in the center of the wood mold, had an aperture slightly wider than the diameter of the bolt, which had a particular cushioning effect during the concrete pouring process. The anchor solid was cured without demolding for 28 days at room temperature. Three standard test blocks were prepared under the same conditions to measure the mechanical indexes of concrete.

3.3.3. Test apparatus

The load was applied by hydraulic jack in the test, and the load sensor and hydraulic jack were calibrated before usage; a multimeter was used to measure the displacement of the anchor rod and anchor solid. Table 3 provides the parameters of the detection device.

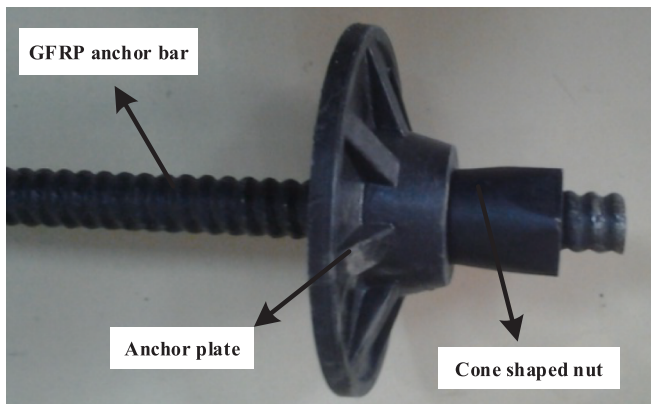


Fig. 2. Physical picture of mechanical anchor.

Table 2

Test anchor parameters.

Test number	Anchor diameter/mm	Anchorage length/mm	Anchoring method	strength grade of concrete	concrete block size/mm
L420	28	420	Linear anchorage	C25	800 × 800 × 500
L840	28	840	Linear anchorage	C25	800 × 800 × 900
M420	28	420	Mechanical anchorage	C25	800 × 800 × 500
M840	28	840	Mechanical anchorage	C25	800 × 800 × 900

Table 3

Parameters of test and detection device.

Category	Quantity/unit	Range/kN or mm	Resolution/kN or mm
Manual hydraulic jack	2	500 kN/20 cm	/
BHR-4 type load sensor	1	500 kN	1.0 kN
Multimeter with magnetic stand	6	30 mm	0.01 mm
YJ-32 digital display instrument	2	/	/

3.4. Test process

In the test preparation stage, the mold was made first, and holes were reserved on the mold for the anchor bars to pass through, then installed the test bolt in the reserved hole, and ensured the accuracy of the anchorage length, for details, saw Fig. 3(b). The test bolt was fixed with steel bars to ensure it was subjected to horizontal forces to prevent the bolt from deflecting due to concrete pouring. While pouring concrete, vibration was applied for a certain period to ensure the anchor solid's compactness, each vibrating duration was 20 to 30 s. After 28 days of curing, the compressive strength of the standard concrete test block was noted as 25.4 MPa, which met the strength requirements.

The anchorage system should be preloaded prior to the test in order to verify the operation of the testing equipment and anchorage components. After calibrating the jack and load sensor, the test loading was started. Two hydraulic jacks were loaded synchronously, and a step-by-step loading method was adopted in the test. At a constant speed of 40 kN per stage, the loading time was kept to within 5 to 8 s, and the load on the subsequent two stages was stable for 15 min until the bolt broke. Following the completion of each loading stage, data were promptly gathered. During load stabilization, measurements and records were taken every 5 min. The test was carried out in strict compliance with the relevant specifications of *Technical Specification for Anti-floating Anchors* (YB/T4659-2018) [27]. The test process is demonstrated in Fig. 3.

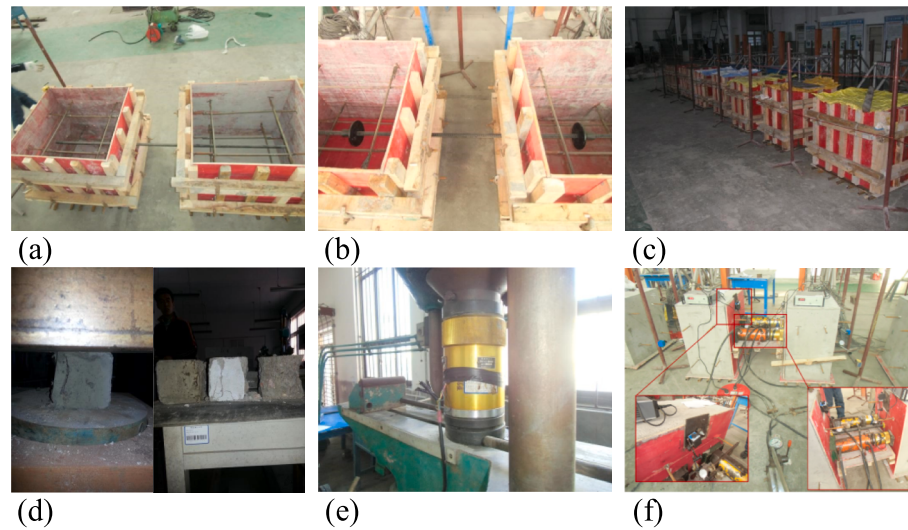


Fig. 3. Photos of the laboratory test site: (a) Make the mould (b) Install the bolt (c) Concrete pouring and curing (d) Concrete strength test (e) Calibration of load sensor (f) Test load.

4. Test results and analysis

4.1. Test failure load and failure mode

The main failure modes of the test bolt were found to be bolt pulling-through failure, followed by bolt fracture failure. The anchors numbered Z420, J420 and J840 were damaged by anchor pulling-through, and the anchors numbered ZM-840 were damaged by anchor fracture.

The appearance of the anchor body and anchor solid did not change during the loading process of specimen Z420 when the test load was low (load 100 kN), and the rate of change of the displacement dial indicator reading was slow as the load increased. When the load was loaded to 120 kN, the specimen issued an intermittent “crackling” sound, and the rate of change of displacement dial indicator reading increased. When the load was loaded to 215.0 kN, the anchor body of the specimen was pulled through. The load digital display instrument showed that the load decreased rapidly, and anchor pulling-through failure occurred in the specimen, as shown in Fig. 4(a). During the loading process of specimen Z840, when the test load was at a medium or low level (load less than 200 kN), the appearance of the anchor body and anchor solid did not exhibit a change. The displacement dial indicator reading observed a low rate of change as the load increased. When the load was further increased, the rate of change of displacement dial indicator reading increased and was accompanied by a continuous “crackling” sound. Since the 356.0 kN load was reached, the specimen broke in a large area, and the anchor body appeared to have significant plastic deformation. The digital load display instrument showed that the load decreased rapidly, and anchor fracture failure occurred in the specimen, as shown

in Fig. 4(b). During the test, the “crackling” sound occurred mainly due to the shear failure of the interface between the outermost fiber of GFRP bars and concrete, and the shear stress of GFRP bars was gradually transferred to the fiber inside through the resin. A pronounced splitting failure mode was not observed in the tests, even though such type of failure can become very relevant depending on the concrete confinement situation (see e.g. Randl et al. [28]). The pulling failure of specimens J420 and J840 was similar to specimen Z420. The test failure load, displacement, and failure mode of the test anchor are depicted in Table 4.

As shown in Table 4, the bolt’s failure load increased with the anchorage length compared with the same anchorage form. Compared with the same anchorage length, the failure load of the bolt with mechanical anchorage was noted as 1.08–1.24 times larger than that with

Table 4
Test results of GFRP anti-floating anchor.

Anchor number	Anchoring method	Anchorage length/mm	Breaking load/kN	Left/right displacement/mm	Failure mode
Z420	Linear anchorage	420	215.0	1.69/4.24	pull-out failure
Z840	Linear anchorage	840	356.0	7.54/7.66	fracture failure
J420	Mechanical anchorage	420	267.2	4.54/5.13	pull-out failure
J840	Mechanical anchorage	840	384.4	8.25/8.98	pull-out failure

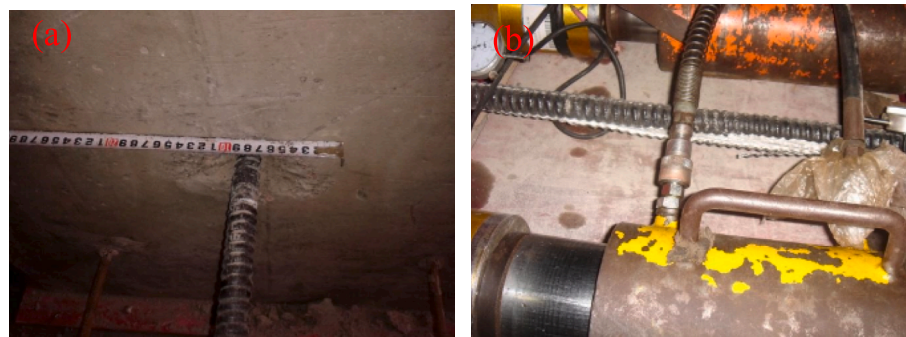


Fig. 4. Anchor failure phenomenon: (a) pull-out failure (b) fracture failure.

linear anchorage. Therefore, the mechanical anchorage could effectively improve the GFRP anti-floating anchor's bearing capability.

During the test, the anchor of Z420, J420, and J840 specimens exhibited pull-out failure, indicating that the bonding force between the anchor bolt and the anchor solid was less than the ultimate bearing capacity of the bolt body and the strength of the anchor did not fully develop. According to the stipulations of anchorage length in China's *Code for Design of Concrete Structures* (GB 50010–2010) [29], as shown in Eq. (1), in this test, C25 concrete was used to anchor GFRP bolt with a diameter of 28 mm, the tensile strength of GFRP was noted as 702 MPa, the tensile strength of concrete was 2.54 MPa, and the rib parameter of the anchor was 0.14 according to [29]. By calculation, the basic anchorage length was found to be 1.08 m, which further validated the test results.

$$l_a = \alpha \frac{f_y}{f_t} d \quad (1)$$

where l_a is the designed anchorage length of the anchor; α depicts the rib parameter of the anti-floating anchor; f_y means the tensile strength of the anchor; f_t depicts the tensile strength of concrete; d is the diameter of the anchor.

During the test loading process, a “crackling” sound was detected from the GFRP anchor. The reasons behind this behavior were as follows: At the beginning of the test loading, the glass fiber of the GFRP anchor and the resin matrix bore the load together. The resin matrix reached the load-bearing limit with the gradual load increase. Since the load further increased beyond the ultimate bearing capacity of the resin matrix, at this time, the additional load increment was borne by the glass fiber of the GFRP anchor. When the test load reached the GFRP anchor's bearing limit, the glass fiber fractured or lost its bond with the resin matrix, which reduced the bolt's bearing capacity. The consistency of the bonding strength between the resin matrix and the glass fiber, however, could not be ensured during the manufacturing of the bolt. Therefore, the GFRP anchor would produce different degrees of damage during loading, which was reflected in the “crackling” sound during loading.

4.2. Load-displacement relation of anti-floating bolt

The load–displacement curve can directly demonstrate the change rule of the pulling load and the bolt displacement under the pulling load's action. In this test, a dial gauge was installed on the anchor solid at a specific distance from the anchor body, monitoring the relative displacement between the anchor solid and the anchor body. The measured displacement was the sum of the anchor's elastic–plastic deformation and the anchor's relative displacement to the anchor solid. Fig. 5 depicts the load–displacement curve relationship of the test bolt.

The load–displacement curve was separated into two stages, as

shown in Fig. 5, from zero loading to failure of the test bolt. In the relaxation section, the displacement of the anchor relative to the anchor solid increased slowly with the increase of the load. In the acceleration section, the displacement of the anchor relative to the anchor solid increased with the increase of the load. There was an inflection point between the relaxation and the accelerating sections, and there was no apparent backward shift with the increase of the anchorage length. The inflection point of specimen Z420 and specimen J420 appeared at a load of 200 kN, and the load borne by the relaxation section accounted for 95% and 75% of the ultimate load, respectively, while the relative displacement of the bolt in the relaxation section accounted for 57% and 45% of the total displacement. The inflection point of specimens Z840 and J840 also appeared at a load of 200 kN, and the load borne by the relaxation section accounted for 56% and 52% of the ultimate load, respectively, while the relative displacement of the anchor rod in the relaxation section accounted for 28% and 23% of the total displacement. The results demonstrated that using the same anchorage method, the relative displacement and proportion of load of the anchor bolt in the relaxation section decreased noticeably as the anchorage length increased. There was a slight correlation between the value of the inflection point and anchorage length. The load–displacement curve of the bolt with mechanical anchorage was smoother, and its inflection point was not as prominent as that of the linear anchorage, as shown in Fig. 5.

It can be concluded from Fig. 5 that the load–displacement curves of the four sets of test anchors mostly coincided when the load was less than 200 kN, indicating that the displacement of anchors with varying anchorage lengths was the same under this test condition. The reasons were as follows: Under the action of load, the load was transferred from the loading end to the deep anchor solid utilizing axial force, the chemical bonding, friction, and mechanical bite force between anchor and anchor solid, providing interface bonding force and counteract with axial force to achieve the effect of anchorage. With the axial force transferred to the depth of the anchor solid, chemical bonding and friction played a function first; however, this contribution was minor. The axial force increased with increasing test load; when the chemical bonding and friction reached their limits, the surface ribs of the anchor body and the anchor solid had a mechanical bite effect. As the load increased, the mechanical bite caused the concrete-bar interface to break and the mechanical bite to fail, and the axial force was transmitted to the next anchorage section with the same action mechanism as the previous section. When the axial force ran through the bolt, the relative displacement of the bolt appeared as an inflection point; when the axial force ran through the anchor, the relative displacement of the anchor appeared as an inflection point. After that, with the increase of load, the displacement of the anchor increased rapidly, and the failure occurred at the exposed position of anchor bolt.

As shown in Fig. 6, the displacement of specimen Z420 on both sides of the anchor was linearly proportional to the load change during the

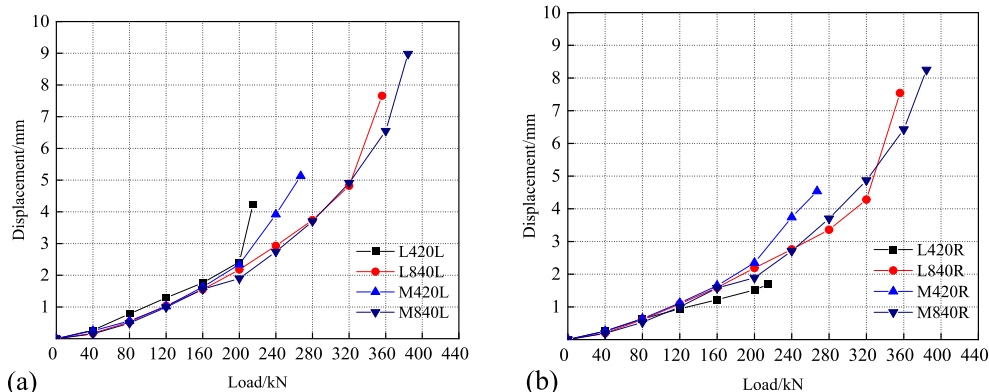


Fig. 5. Load-displacement curve of test bolt (L and R represent left and right side of tensile specimen):(a) Pull test on left anchor (b) Pull test on right anchor.

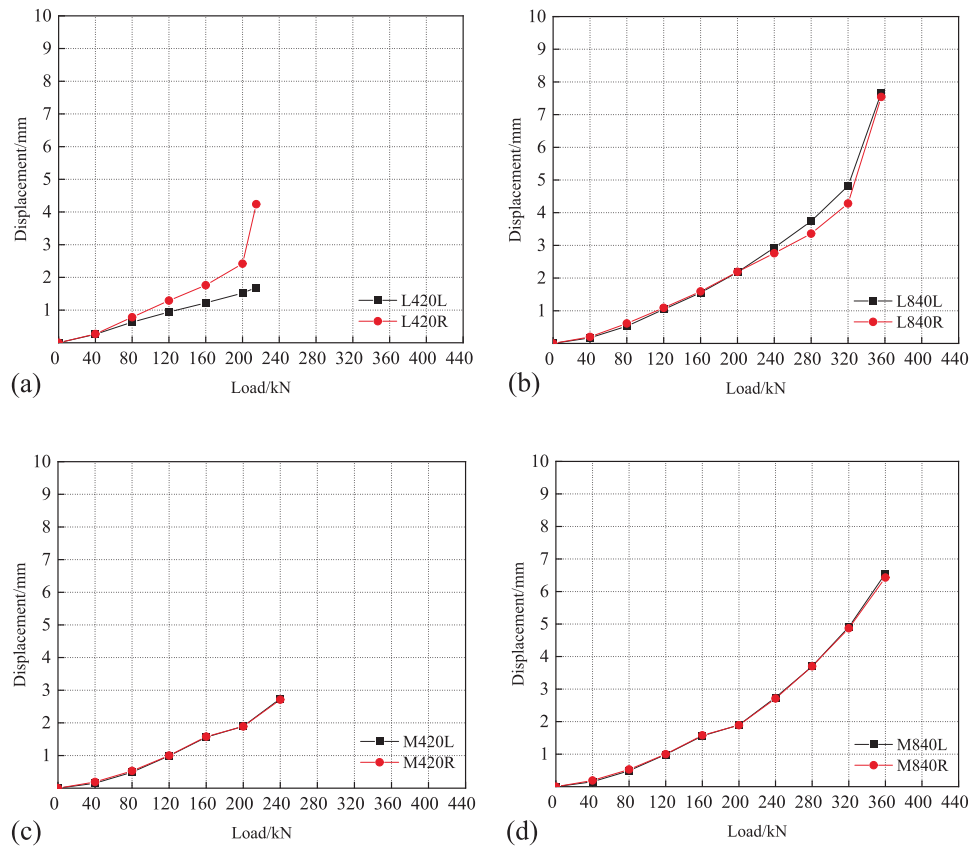


Fig. 6. Load-displacement curve of test bolt: (a) specimen Z420 (b) specimen Z840 (c) specimen J420 (d) specimen J840.

initial loading stage. When the load exceeded 40 kN, the displacement on the left and right sides of the anchor changed significantly. When the load reached 80 kN, the displacement on the left side of Z420 was 0.63 mm, the displacement on the right side was 0.78 mm, and the displacement difference was 0.15 mm. With the increase in load, the displacement difference increased. When the specimen reached the ultimate load of 215 kN, the displacement on the left side of specimen Z420 was 1.69 mm, the displacement on the right side was 4.24 mm, and the displacement difference expanded to 2.55 mm. The reasons investigated were as follows: (i) In the loading process, the installation position of the jack, or the uneven holding surface, caused the eccentric compression of the specimen; the bearing capacity on both sides was different, and thus the displacement had a significant difference. (ii) In the actual construction process, the quality of anchor solid was different, which led to the difference in the bond strength between it and the anchor rod, which led to the difference in the test results. By comparing the load–displacement curves of four groups of tensile specimens, it was found that each group had an overall positive influence on the parallel tensile test. It demonstrated that the test was scientific and the test result was reliable.

4.3. Anchor failure mechanism analysis

4.3.1. Linear anchorage failure mechanism

As shown in Fig. 7, the tension of the anti-floating anchor in the form of linear anchorage was subjected to the pulling force; the pulling force was transmitted to the anchorage depth along the axis direction of the anchor. The anchorage segment of the GFRP bolt was divided into the acting zone and non-acting zone based on the process of transmission. Among them, the action zone was shown as: The bolt had axial force under the action of pulling force, and the relative displacement trend between the bolt and concrete occurred; the chemical bonding, friction,

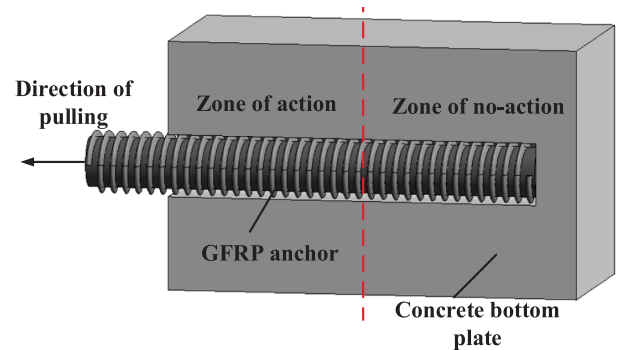


Fig. 7. Schematic diagram of failure mechanism of linear anchorage anchor.

and mechanical bite between interfaces played the bonding role, as the pulling force increased, the stress state of the non-acting zone anchor changed to the acting zone until it ran through the entire anchor solid.

The main bearing mode of the bonding force between the anchor and the concrete floor constantly changed as the axial force of the anchor was transferred to the deep of the anchor solid. As shown in Fig. 8(a), when the axial force of the anchor was small, the relative displacement between the anchor and the concrete floor in the action zone was quite lower, and chemical bonding and friction between the interface played an important role. When the axial force continued to increase, the relative displacement between the anchor body and the concrete floor in the action zone grew, and the anchor rib of the GFRP anchor became entirely occluded with the concrete. The mechanical bite force mainly bore the interface bonding force in this case. The concrete was crushed, and the surface fiber of the GFRP anti-floating anchor fractured when the axial force of the bolt increased, as shown in Fig. 8(b). The non-

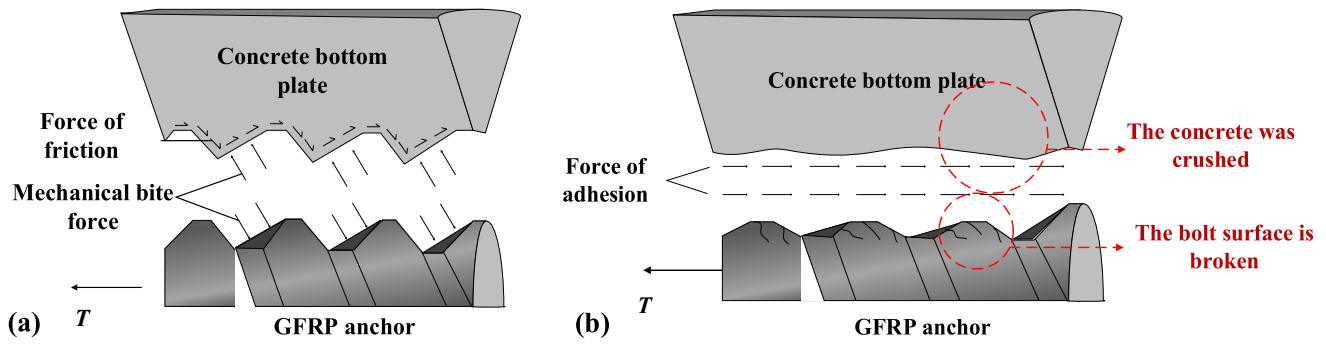


Fig. 8. Meso diagram of failure of linear anchor: (a) Low stress state (b) High stress state.

acting zone eventually transformed into the acting zone, and the force was repeated until the zone of action was filled with anchor solids, at which point the bond force between anchor and concrete could no longer withstand the action of axial force, and failure occurred. Experimental research results conducted by many researchers showed that: In the case of the linear anchor of GFRP anchor, mainly bond slip failure occurred, and unlike metal bolt, the phenomenon of breakage rarely occurred. Because of the material properties of the anchor, mainly because of the weak shear performance of the GFRP anchor, the GFRP anchor is generally at a disadvantage when it occludes the concrete, and the fiber fracture occurs on the surface of the bolt [30].

4.3.2. Mechanism of mechanical anchorage failure

According to the experimental investigation, mechanical anchorage may effectively limit the tensile displacement of the anchor, and its failure process is depicted in Fig. 9. When the anchor was subjected to axial force, the anchoring force was shared by the bonding force between the anchor, the concrete interface, and the anchoring plate bearing pressure. Mechanical anchorage failure can be classified into three stages: In the first stage, when the pulling force was small, the axial force transfer depth was shallow, the anchorage force was borne by the bonding force between the anchor and the concrete interface, and the specific bearing method was the same as linear anchorage. In the second stage, as the pulling force increased continuously, the axial force was transferred to the front end of the anchor plate; however, because the anchor plate limited the anchor's displacement, the interface bonding force of the anchor adjacent to the anchor plate could not be fully exerted; at this time, the anchorage force was mainly balanced by the pressure of the anchor plate. When the pulling force continued to increase, the concrete in the pressure zone of the anchor plate was crushed, and the anchor plate was about to fail. In the third stage, the anchorage plate was damaged by further increasing axial force. The bearing pressure of the anchor plate was reduced rapidly, and the bonding force at the interface adjacent to the anchor plate continued to play a role until the anchor slip failure occurred. In the process of anchor failure, the anchor plate's design was imperfect and could not fully meet

the engineering requirements. The anchor plate at the anchor location was prone to fracture due to shear force, and this part was called the vulnerable area of the bolt. In the future research, the design of the anchor plate in the mechanical anchor is open to improvement.

5. Mechanical anchoring efficiency analysis

When the base plate is thin, and the bolt cannot be bent, adding the anchor plate increases the bonding force between the anchor and the foundation plate. The commonly used anchoring methods mainly include support type, clip type, gripping type, and so on. The basic performance of the anchorage plate, including strength, fatigue resistance, force transmission and shrinkage of anchor zone, has to meet the relevant provisions of *Mechanical anchors for use in concrete* (JGT 160–2017) [31] and *Anchorage, grip, and coupler for prestressing tendons* (GB/T 14370–2015) [32]. Among them, the efficiency coefficient of the anchor measured by static load test is not less than 0.95, and the total strain when the ultimate tension is reached is not less than 2.0%, calculated according to Eq. (2):

$$\eta_a = \frac{F_{apu}}{\eta_p \cdot F_{pm}} \quad (2)$$

where F_{apu} is the measured value of ultimate tension, unit: kN; η_p depicts the anchorage coefficient of reinforcement, generally 1.0 for non-metallic reinforcement; F_{pm} means the measured average value of ultimate tensile strength of reinforcement, unit: kN.

The anchoring efficiency coefficient in Eq. (2) was used to describe the anchoring efficiency of mechanical anchors in the test to measure their anchoring efficiency. The efficiency coefficient of test anchor J420 was 0.62 after calculation, and the efficiency coefficient of test anchor J840 was 0.89. The anchorage coefficient increased with the increase in anchorage length. The anchorage device material used in the test was the same as that of the anchor, which was GFRP material. Due to the anisotropy of the material itself and the limited shear strength, its anchoring efficiency could not completely meet the code's requirements.

6. Effect of anchorage length on anchorage performance

To explore the effect of the anchorage length of the GFRP bolt on anchoring performance, the mechanical anchorage length was converted into the linear anchorage length. In order to better simulate the underground anti-floating environment, according to the *Design Code for Concrete Structures of Port and Waterway Engineering* (JTS 151–2011) [33] the anchoring effect of a GFRP anchor with linear anchoring is 0.7 times that of a mechanical anchoring anchor with the same anchoring length [33], converting the anchoring length of the test bolt. The specific value of the conversion is shown in Table 5. The curve of the converted value of anchorage length and failure load was created, as shown in Fig. 10.

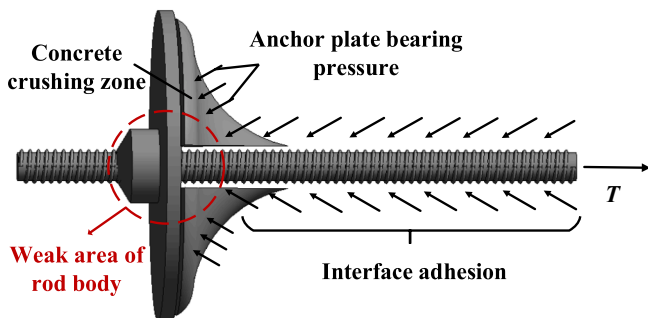


Fig. 9. Schematic diagram of mechanical anchorage bolt failure mechanism.

Table 5
Statistical table of GFRP anti-floating anchor test.

Anchor number	Anchoring method	Anchorage length/mm	Anchor length conversion value/mm	Breaking load/kN
Z420	Linear anchorage	420	420	215.0
J420	Mechanical anchorage	420	600	267.2
Z840	Linear anchorage	840	840	356.0
J840	Mechanical anchorage	840	1200	384.4

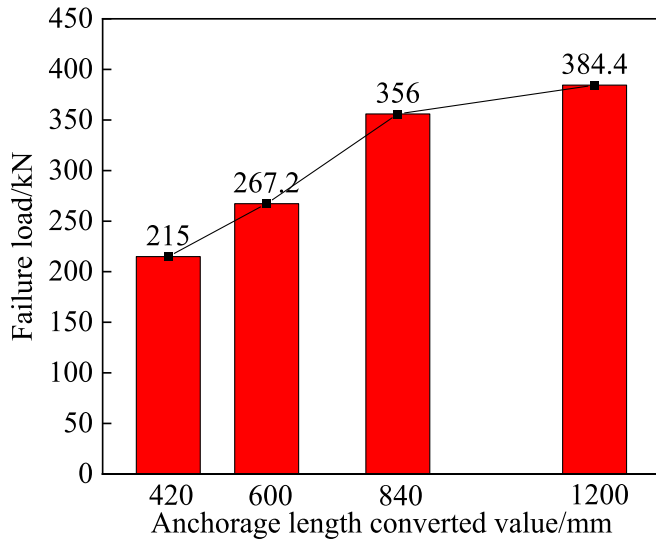


Fig. 10. Relationship between the converted value of anchorage length and failure load.

As can be seen from Table 5 and Fig. 10, the failure load of the GFRP non-floating anchor increases with increasing anchor length, confirming that increasing anchor length can improve bearing capacity. It can also be found from Fig. 10 that when the anchorage length exceeds 840 mm, the effect of increasing the anchorage length to improve the bearing capacity of the anti-floating anchor is not obvious. Therefore, it can be concluded that when the anchor length is approximately 840 mm in this test condition, there is a limit value of anchorage.

7. Anchorage length calculation theory

7.1. Calculation method of anchorage length

Laboratory anchorage experiments of GFRP anchors with different anchorage lengths reveal that the anchorage performance between GFRP anchors and the concrete floor is closely related to the anchorage length [34,35]. Under certain limits, the anchorage performance improves with the increase of the anchorage length. Due to the lack of a theoretical calculation method for anchoring between GFRP bolt and concrete floor, this paper aimed to explore a more theoretical calculation method of fitting engineering by comparing the specifications of anchoring length in different countries.

According to the *Code for Design of Concrete Structures* (GB 50010–2010) [29] of China, the design requirements for the anchorage length of reinforced Concrete are shown in Eq. (1).

According to the requirement of the American Standard *Building code Requirement for Structural Concrete* (ACI 318–11) [36] (hereinafter referred to as ACI 318) for anchorage length, the calculation is shown in

Eq. (3).

$$l_a = \frac{a\psi\lambda f_y d}{\sqrt{f_c}} \quad (3)$$

where a represents the anchoring mode, 0.075 for the linear anchor, 0.02 for the curved anchor, and 0.016 for the mechanical anchor; ψ means the coating parameter, which is 1.2. λ is the concrete parameter, 1.0 for ordinary concrete; f_c depicts the compressive strength of concrete.

In accordance with the European Standard *Eurocode 2, Design of Concrete Structures* (BS EN 1992–1) [37] (hereinafter referred to as BS EN 1992), the linear anchorage length is calculated according to Eq. (4), with the bending anchorage length being l_b minus bending length.

$$l_b = \frac{d\sigma_{sd}}{9\eta_1\eta_2f_t} \quad (4)$$

where σ_{sd} is the actual stress of anchor; η_1 is the bond state coefficient of anchor, set at 0.7; η_2 is the coefficient related to the anchor, when $d < 32$, $\eta_2 = 1.0$; f_t is the tensile strength of concrete.

According to *Australian Standard Concrete Structure* (AS 3600–2009) [38], the calculation method of anchor anchorage length is shown in Eq. (5), in which the influence of various parameters on anchor anchorage length is considered.

$$l_b = 0.5k_4k_5 \frac{0.2k_1k_3f_yd_b}{k_2\sqrt{f_c}} \quad (5a)$$

$$k_2 = \frac{132 - d_b}{100} \quad (5b)$$

where k_1 , k_2 , k_3 , k_4 , and k_5 take into account the influence coefficients of concrete protective layer thickness, anchor diameter, concrete type, reinforcement placement position, and stress state, respectively. According to the test parameters in this paper, $k_1 = k_4 = k_5 = 1.0$, and $k_3 = 1.3$.

According to the Japanese standard, *JSCE Standard Specifications for Concrete structure-2012, Design* (JSCE 2012) [39], the linear anchorage length is calculated according to Eq. (6). The bending anchorage length is the basic anchorage length l_b minus 10d (d is the sectional diameter of the anchor, and the same as below).

$$l_b = \alpha \frac{f_y}{4f_b} d \quad (6)$$

where α is the rib parameter of the anti-floating anchor, with the value ranging from 0.6 to 1.0, f_b is concrete bonding strength.

7.2. Comparative analysis of anchorage length calculation

Currently, there are standards, including Chinese standard (GB 50010-2015), American standard (ACI 318-11), European standard (BS EN 1992), Australian standard (AS 3600-2009), and Japanese standard (JSCE 2012) for calculating the anchorage length of the anchor. Since there is no uniform international standard, there are some variances in the regulations of anchorage length in different countries. The relevant anchorage length standards in several nations were calculated using the test parameters of this laboratory test. The calculation process and results are shown in Table 6.

It can be seen through calculation that the calculation results based on Chinese, European, and Japanese standards are 2–3 times the calculation results of American and Australian standards, as shown in Table 6. It can be seen from the calculation formula that the American standard and the Australian standard take the concrete compressive strength as the influencing factor of anchorage length. In contrast, the Chinese standard takes the concrete tensile strength as the influencing factor of anchorage length. The formulas for calculating anchorage

Table 6

Calculation results of theoretical anchorage length of concrete in various countries.

Calculation basis	Theoretical anchorage length calculation process	Calculated anchorage length /mm
China standard	$l = 0.14 \times 702\text{MPa} / 2.54\text{MPa} \times 28\text{mm}$	1080
American standard	$l = 0.075 \times 1.2 \times 1.0 \times 702\text{MPa} / \sqrt{25.4\text{MPa}} \times 28\text{mm}$	351
European standard	$l = 496\text{MPa} / (2.54\text{MPa} \times 9 \times 0.7 \times 1.0) \times 28\text{mm}$	868
Australian Standard	$l = 0.5 \times 0.2 \times 1.3 \times 702\text{MPa} / (\sqrt{25.4\text{MPa}} \times (134 - 28) / 100) \times 28\text{mm}$	488
Japanese Standard	$l = 0.6 \times 702\text{MPa} / (5.32\text{MPa} \times 4) \times 28\text{mm}$	554

length in the Japanese and European standards are the same as those in the Chinese standard form. However, they take the bond strength of concrete and the actual stress of the anchor as parameters, respectively. Yet, it is difficult to obtain the bond strength and actual stress of concrete in engineering design. According to the calculation results of the Japanese standard, it can be seen that the calculated value of anchorage length is small (the anchor length is shorter). Compared with the test results, when the anchorage length is 554 mm, the strength utilization ratio of the GFRP anchor is low. In conclusion, when designing GFRP anti-floating anchor, the anchoring length of the anti-floating anchor bolt can be calculated according to Chinese and European standards. It can be concluded that the calculation method of the Chinese standard is more conservative.

8. Conclusion

- (1) The failure modes of the bolt in the laboratory model test are a pull-out failure and fracture failure. The former originated because the bond force between the anchor and concrete floor is less than the ultimate pulling load of the anchor, the latter because the bond force between the anchor and concrete floor is greater than the ultimate pulling load of the anchor, and the failure phenomenon is obvious. Compared with the same anchorage method, the anchor's failure load increases with the anchorage length. The failure load of the anchor with mechanical anchorage is 1.08–1.24 times greater than that with linear anchorage for the same anchorage length. Mechanical anchorage can boost the bearing capacity of GFRP anti-floating anchors.
- (2) The anchor's load–displacement curve is separated into two sections: relaxation and acceleration section. The relative displacement and load proportion of the anchor in the relaxation section decreases noticeably as the anchorage length increases using the same anchoring method. The inflection point value has little correlation with the anchorage length value. The load–displacement curve of the mechanical anchor is smoother, and its inflection point is less noticeable than that of the linear anchor.
- (3) The anchoring coefficient increases with the length of the anchorage, according to an analysis of the anchoring efficiency of the anchorage device utilized in the test. The failure mechanism of linear and mechanical anchorage is summarized from the *meso*-view, and the shear failure law of the interface between the anchor and concrete is directly expressed as the load is gradually transferred to the deep along the axis of the anchor.
- (4) According to the Chinese concrete code, mechanical anchorage length is converted into linear anchorage length, demonstrating that the ultimate pulling load of the GFRP anti-floating anchor increases with anchorage length, and it is inferred that the ultimate anchorage length of the GFRP anchor is about 840 mm

under test conditions. According to the calculation results of the relevant standards of the basic anchoring length of the anti-floating anchor in various countries, it is suggested to use the Chinese and European standards to calculate the basic anchoring length of the GFRP anti-floating anchor.

CRediT authorship contribution statement

Gan Sun: Writing – original draft. **Nan Yan:** . **Xiaoyu Bai:** Writing – review & editing. **Junwei Liu:** Investigation. **Dongshuai Hou:** Writing – review & editing. **Songkui Sang:** Data curation. **Mingyi Zhang:** Conceptualization. **Pan Wang:** Writing – review & editing. **Desheng Jing:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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